INCH-POUND
A-A-59403B
22 October 2019
SUPERSEDING
A-A-59403A
January 30, 2008

COMMERCIAL ITEM DESCRIPTION

CLOTH, SEAT COVER, POLYESTER OR NYLON, WATER-RESISTANT

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE.

1.1 Scope. This commercial item description covers three types of UV-stabilized polyester water-resistant cloth. The cloth is intended for use in the manufacture of seat covers, tops and tarps for motorized equipment.

2. CLASSIFICATION.

2.1 Classification. The cloth shall be of the following type:

Type I	-	Cloth, Uncoated, Polyester or Nylon, Woven
Type II	-	Heavy Duty Cloth, Uncoated, Polyester or Nylon, Woven
Type III	-	Highly Water Resistant, Heavy Duty Cloth, Uncoated, Polyester or Nylon, Woven
Type IV	-	Cloth, Heavy Duty, Polyester, Coated, Flame Resistant, Woven

3. SALIENT CHARACTERISTICS.

3.1 Cloth. All cloth shall consist of high-tenacity, either bright, semi-dull, or clear, filament polyester yarn. All types shall be woven and the weave shall be plain. The cloth shall conform to one of the types described in Table I and shall conform to the requirements specified in 3.1 through 3.9.

3.2 Weight. The weight of all cloth shall be 12 + 3 ounces per yard (oz/yd2) when measured IAW ASTM D3776/D3776M, Option C.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be emailed to usarmy.detroit.rdecom.mbx.tardec-standardization@mail.mil

AMSC N/A FSC 25GP

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

3.3 Minimum flame resistance. Flames shall progress no more than 4.0 inch per minute (inch/min) when tested IAW FMVSS 302. Type IV shall provide flame resistance of less than 6.0 inch char length and less than 3 seconds after flame when tested to ASTM D 6413/D6413M vertical burn test.

3.4 Operational temperature range required. The operational temperature range required shall be -50 to 150 degrees Fahrenheit (°F) when tested IAW 3.4.1 and 3.4.2.

TABLE I. Cloth description and verification.

Characteristics	Type I	Type II	Type III	Type IV	Test
Tensile strength – minimum	260 lbs Warp 175 lbs Fill 1/	450 lbs Warp 400 lbs Fill	450 lbs Warp 400 lbs Fill	450 lbs Warp 400 lbs Fill	ASTM D5034 2/
Tear strength	10 lbs Warp 11 lbs Fill	45 lbs Warp 40 lbs Fill	45 lbs Warp 40 lbs Fill	45 lbs Warp 40 lbs Fill	ASTM D2261
Elastic stretch range	3-7% Warp 11-18% Fill	3-11% Warp 3-11% Fill	3-11% Warp 3-11% Fill	3-11% Warp 3-11% Fill	SAE J855
Abrasion resistance (minimum)	1000 cycles	2000 cycles	2000 cycles	2000 cycles	ASTM D3884 3/
Tensile strength retention after UV exposure minimum	104 lbs Warp 70 lbs Fill	230 lbs Warp 160 lbs Fill	175 lbs Warp 125 lbs Fill	175 lbs Warp 125 lbs Fill	SAE J-2412 & ASTM D5034 4/
Tensile strength retention after wear degradation maximum	170 lbs Warp 150 lbs Fill	400 lbs Warp 350 lbs Fill	400 lbs Warp 350 lbs Fill	400 lbs Warp 350 lbs Fill	ASTM D4157 & ASTM D5034 5/
Air permeability ($ft^3/min/ft^2$) 6/ minimum	1.3	1.3	0.5	1.3	ASTM D737
Water resistance (cm water in column) -- minimum	10	10	30	10	AATCC 127 7/

NOTES:

1/ lbs = pounds.

2/ Tensile strength. A constant rate-of-extension (CRE) testing machine shall be used.

3/ There shall be no visible wear-through when Taber-tested with a 1 kilogram (kg) weight and a CS-10 wheel for 1000 cycles for type I and 2000 cycles for types II, III, and IV.

4/ Strength retention after UV exposure. After being exposed to a xenon-arc lamp at 225.6 kilojoules per square meter (kj/m2) In Accordance With (IAW) SAE J2412, cloth shall retain specified tensile strength IAW ASTM D5034.

5/ Strength retention after wear degradation. Cloth shall retain specified tensile strength after 100,000 double rubs (Wyzenbeek abrasion) using a #8 Duck as the abrader IAW ASTM D4157. Tensile strength IAW ASTM D-5034.

6/ ft3/min/ft2 = Cubic feet per minute per square foot.

7/ Water resistance. Water shall not pass through the cloth when a column of the minimum height specified is used. Testing IAW AATCC 127.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

3.4.1 Resistance to low temperature test. A 1 x 4-inch specimen of the cloth with the long dimension warp-wise, and a 1 x 4-inch specimen of the cloth with the long dimension filling-wise, shall be exposed for 4 hours at a temperature of -50 ± 5°F. The sample, still in the test atmosphere, shall be bent sharply 180 degrees over a 1/8 inch steel rod exposed in the test chamber with the test specimen. The cloth shall not crack or flake.

3.4.2 Resistance to high temperature test. A 2 x 6 inch specimen of the cloth shall be exposed for a period of 6 hours in an oven maintained at a temperature of +150° ± 2°F. At the end of this period, the specimen, still maintained at the test atmosphere shall be bent sharply 180 degrees over a 1/8 in steel rod. The cloth shall show no evidence of tackiness, blistering, or softening.

3.5 Water wicking (minimum). The minimum water wicking shall be 3.0 in the ISO Gray scale unit of measurement when tested IAW SAE J913, option C.

3.6 Dimensional stability (maximum shrinkage). The maximum shrinkage shall be 3.5% in the warp direction and 2.5% in the fill direction when tested IAW SAE J883.

3.7 Mildew resistance (rating). The mildew resistance shall be 0 in the ISO Gray scale unit of measurement when tested IAW ASTM G21.

3.8 Color. The color of the finished cloth shall match green (34094) or tan camo (33446) of SAE AMS-STD-595 or be specified in the contract for a color specified. There shall be no light areas or windows due to absence or poor blending of pigmentation. Visual testing is to be done on a 4 x 20 inch swatch of cloth with a light of at least 538 lux at the working surface. The color of the finished cloth shall meet the spectral reflectance properties in Tables II, III, and IV.

3.9 Length and put-up. The cloth shall be put-up in full width rolls in lengths as required by the contract or purchase order. The total length shall be marked on an identifying document. The ends of the pieces shall be overlapped and not joined by a seam. When unrolled, there shall be no rolled, curled, folded, doubled, scalloped, or wavy edges, which prevent a flat lay of the cloth.

TABLE II. Spectral reflectance values (percent) for green (34094).

Wavelength, Nanometers	Reflectance		Wavelength, Nanometers	Reflectance	
	Min.	Max.		Min.	Max.
600	3	12	740	7	52
620	3	12	760	11	60
640	3	12	780	17	64
660	3	13	800	24	70
680	3	15	820	32	80
700	3	28	840	37	86
720	5	40	860	49	88

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

TABLE III. Spectral reflectance values (percent) for tan camo (33446).

Wavelength, Nanometers	Reflectance		Wavelength, Nanometers	Reflectance	
	Min.	Max.		Min.	Max.
700	45	65	800	45	65
720	45	65	820	45	65
740	45	65	840	45	65
760	45	65	860	45	65
780	45	65			

TABLE IV. Spectral reflectance values (percent) for specified colors.

Wavelength	Desert Sand 23525		Urban Gray 26260		Foliage Green 24165	
Nanometers (nm)	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
600	28	40	12	26	8	18
620	30	42	14	26	8	18
640	34	48	14	28	8	20
660	38	56	14	30	10	26
680	44	60	18	34	10	26
700	46	66	24	38	12	28
720	48	68	26	42	16	30
740	48	72	30	46	16	30
760	50	74	32	48	18	32
780	54	76	34	48	18	34
800	54	76	34	50	20	36
820	54	76	36	54	22	38
840	56	78	38	54	24	40
860	56	78	40	56	26	42

4. REGULATORY REQUIREMENTS.

4.1 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, IAW paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE.

5.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

5.2 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market expect for any changes necessary to conform to this commercial item description. The Government reserves the right to require proof of such conformance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

5.3 Market acceptability. The offeror shall prepare written certification that the cloth provided has been a stable production item for at least three years or is the basis for current orders if a new product. The use of the term “commercial item” in this document does not imply any items offered are not required to conform to all requirements specified herein.

5.4 Standard test conditions. Physical tests shall be conducted with both the specimen and test apparatus under standard conditions as defined in ASTM D618.

6. PACKAGING.

6.1 Packaging. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

7.1 Ordering data. Acquisition documents must specify the following:

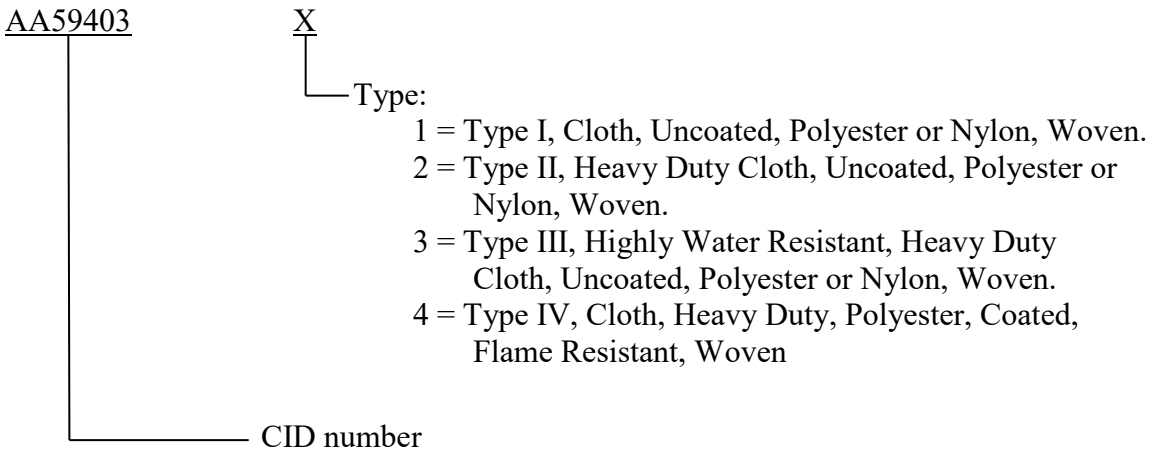
a. Title, number, and date of this CID.

b. Length/width of rolls.

c. Level of preservation, packaging, and packing required.

d. Color required.

7.2 Part Identification Number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

7.3 Address for obtaining copies of reference documents.

7.3.1 Government publications. The Code of Federal Regulations (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

a. Copies of AATCC 127, "Water Resistance: Hydrostatic Pressure Test" are available online at http://www.aatcc.org or from The American Association of Textile Chemists and Colorists (AATCC) Standards, 1 Davis Drive, P.O. Box 12215, Research Triangle Park, NC 27709-2215.

b. Copies of ASTM D3776/D3776M, "Standard Test Method for Mass Per Unit Area (Weight) of Fabric"; ASTM D2261, "Standard Test Method for Tearing Strength of Fabrics by the Tongue (Single Rip) Procedure (Constant-Rate-of-Extension Tensile Testing Machine"; ASTM D3884, "Standard Guide for Abrasion Resistance of Textile Fabrics (Rotary Platform, Double-Head Method)"; ASTM D4157, "Standard Test Method for Abrasion Resistance of Textile Fabrics (Oscillatory Cylinder Method)"; ASTM D5034, "Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)"' ASTM D6413/D6413M, "Standard Test Method for Flame Resistance of Textiles (Vertical Test)", ASTM D737, "Standard Test Method for Air Permeability of Textile Fabrics"; ASTM D618 "Standard Practice for Conditioning Plastics for Testing"; and ASTM G21 "Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi" are available online at http://www.astm.org or from the ASTM International, 100 Bar Harbor Drive, West Conshohocken, PA 19428-2959.

c. Copies of FMVSS 302, "Flammability of Interior Materials-Passenger Cars, Multipurpose Passenger Vehicles, Trucks and Busses" are available online at www.fmvss.com or from the U.S. Department of Transportation, National Highway Traffic Safety Administration, Office of Vehicle Safety Compliance, 400 Seventh Street SW, Room 6115, Washington, DC 205090.

d. Copies of SAE J855, "Test Method of Stretch and Set of Textiles and Plastics"; SAE J883, "Test Method for Determining Dimensional Stability of Automotive Textile Materials"; SAE J2412, "Accelerated Exposure of Automotive Interior Trim Components Using a Controlled Irradiance Xenon-Arc Apparatus"; and SAE J913, "Test Method for Wicking of Automotive Fabrics and Fibrous Materials" are available online at http://sae.org or from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-0001.

e. Copies of SAE AMS-STD-595, "Colors Used in Government Procurement" are available online at http://sae.org.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.

Custodians:
Army – AT
Navy - NU
Air Force - 99

Review Activities:
Army - GL, MI
Navy – MC
DLA – CT
Air Force - 84

Preparing Activity:
Army – AT

(Project 25GP-2019-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:13Z
Check the source to verify that this is the current version before use.